Roy Yao

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EDUCATION:

University of California, Berkeley **(Aug 2023- May 2027)**

B.S. Electrical Engineering & Computer Science, Bioengineering, GPA: 3.84

HONORS/AWARDS: Dean's List, Bioengineering Honor Society, Rose Hill Summer Scholarship, GLOBE Discovery Scholarship

Relevant Coursework: CS 189: Machine Learning, **CS 162:** Operating Systems and System Programming, **EE 120:** Signals and Systems; **EECS 127:** Optimization Models in Engineering; **CS 170:** Efficient Algorithms and Intractable Problems; **EE 145B:** Medical Imaging Signals and Systems; **BioE 101:** Instrumentation in Biology and Medicine; **BioE 121, BioE 121L:** BioMEMS, Medical Devices, and BioNanotechnology Laboratory; **EECS 16B:** Circuits & Devices; **CS 61A:** Structure and Interpretation of Computer Programs; **CS 61B:** Data Structures; **CS 61C:** Computer Architecture/Machine Structures; **CS 70:** Discrete Mathematics and Probability Theory;

DEPARTMENTAL SERVICE & LEADERSHIP:

Webmaster, *Bioengineering Honor Society (BioEHS)*, UC Berkeley **(Jan 2024 -)**

- Ran the BioEHS website and candidate portal, using Django (Python) for backend and React, JavaScript, HTML, and CSS for frontend.
- Created the first interactive course map and technical topic page for bioengineering undergraduates using HTML and CSS.

Co-Vice President, *Bioengineering Mentorship Program (BMP)*, UC Berkeley **(Aug 2024 -)**

- Co-led a student mentorship program for bioengineering undergraduates, supporting mentor-mentee engagement, organizing outreach and programming, and fostering academic and professional community.

EXPERIENCES:

Imaging Scientist Intern, *Varex Imaging*, San Jose **(June 2026 -)**

- Photon-Counting Detectors

Undergraduate Researcher, *Conolly Lab*, UC Berkeley **(Aug 2025 -)**

- Verified, designed, and optimized a low-noise analog front-end circuit for Magnetic Particle Imaging (MPI), including LTspice simulation and PCB implementation of a low-noise preamplifier, achieving a 10 times increase in signal-to-noise ratio (SNR) with 5 MHz bandwidth.

Summer Research Intern, *Chen Lab*, National Tsing Hua University **(Jun 2025 - Jan 2026)**

- Participated in the 2025 Summer Research Program. Contributed to optical setup design, microfluidics experimentation, and algorithm development. Hands-on in SolidWorks, microfluidic systems, pumping systems, CNC, laser cutting, FEB bonding, and PDMS fabrication.
- Developed signal processing and correlation algorithms (bandpass filtering, lag alignment, amplitude thresholding) to validate nanoparticle transit events against lock-in amplifier outputs. Built computer vision pipelines for background subtraction and annular pattern detection, enabling video-based particle counting independent of photodiode DAQ systems.

Software Team Member, *Neurotech@Berkeley*, UC Berkeley **(Jan 2025 -)**

- Utilized a 1D CNN model and data augmentation to achieve 20x accuracy using speckle data. Applied the Latent Consistency Model (LCM) to reduce runtime and enhance the clarity and reduce the runtime of reconstructed fMRI images.
- Maintained and updated the Neurotech@Berkeley official website with HTML/CSS and GitHub-based workflows.

Undergraduate Researcher, *Ganguly Lab*, UC San Francisco **(Oct 2023 - May 2024)**

- Researched under Dr. Karunesh Ganguly's lab. Journal clubbed brain-computer interface (BCI) papers weekly. Coded for empirical mode decomposition (EMD) and channel dropout data argumentation on BCI data. Wrote and applied neural network models such as multilayer perceptron (MLP), convolutional neural network (CNN), recurrent neural network (RNN), and long short-term memory (LSTM) to increase accuracy on BCI action classification.

Summer Researcher, UC Irvine **(Jul 2022 - Aug 2022)**

- Graphed and analyzed in-time physiological signals with OpenBCI. Learned and applied brain imaging modalities. Processed data training program on OpenVibe (brain-computer interface) and P300 Speller (brainwave speller) to spell words with electroencephalography (EEG). Developed a letter-based proximity algorithm to increase the accuracy of the speller using MATLAB.

PROJECTS:

EMG-Based Word Detection Sensor, UC Berkeley **(Apr 2025 - May 2025)**

- Designed and prototyped a facial EMG sensor using Ag/AgCl electrodes, AD620 instrumentation amplifiers, and Arduino MKR Zero ADC for real-time speech-related signal acquisition.
- Implemented two-stage band-pass filtering (20–500 Hz, gain ≈ 4000) and autocorrelation-based classification augmented with cosine similarity, MSE, and CNN algorithms, achieving 99% word detection accuracy.

Silent Speech Decoding, UC Berkeley **(Aug 2025 - Dec 2025)**

- Developed a multimodal ML pipeline for silent-speech recognition using the RVTALL dataset, extracting features from audio, lip landmarks, RGB/depth video, and laser speckle while evaluating cross-modal synchronization. Trained a 1D CNN on augmented laser-speckle time-series data, improving 30-class classification of vowels, words, and sentences to 71.15% test accuracy with 0.71 F1

SKILLS:

- **Programming:** Python, MATLAB, C, Java, SQL
- **Libraries:** NumPy, PyTorch, OpenCV
- **Tools:** Git, LTspice, KiCad, COMSOL, SolidWorks, Arduino
- **Technical Areas:** Signal processing, machine learning, computer vision, circuit design, optimization